Speech-based Natural Language Interface for UAV Trajectory Generation

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   - Mouse Interface
4. Results
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   - Subjective Analysis
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6. Future Work

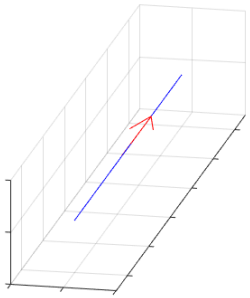
Objective

1. Increasing ubiquity and power of Unmanned Aerial Vehicles (UAVs) makes them compelling tools
   - Data collection tools
   - Carry out repetitive/demanding tasks
2. Use is limited due to complexity of interface
   - Subject matter experts required for operation
   - Significant training
3. Reducing the complexity of UAV interfaces → increased use and expanded user base
   - Intuitive, natural language interfaces

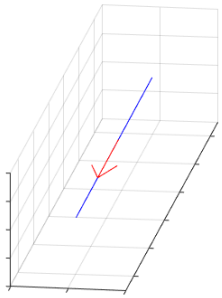
This study examines the use of a speech-based natural language interface compared to a mouse-based interface for UAV trajectory generation

Trajectory Segment Library

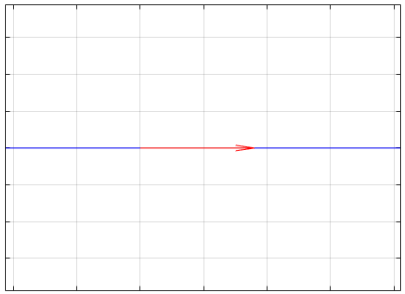
Forward



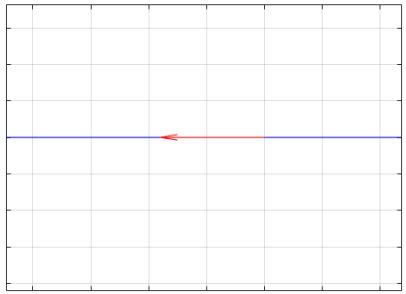
Backward



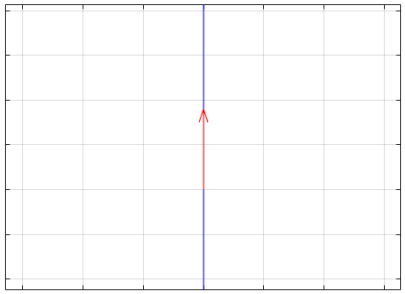
Right



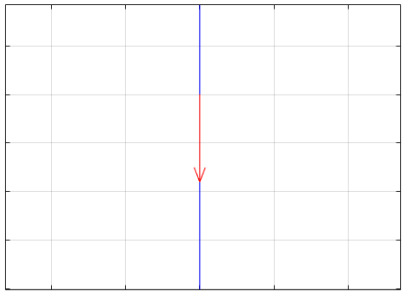
Left



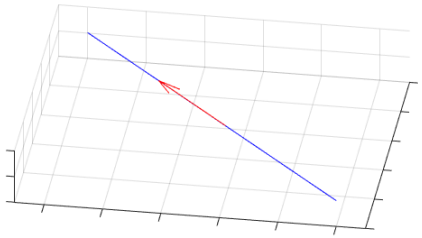
Up



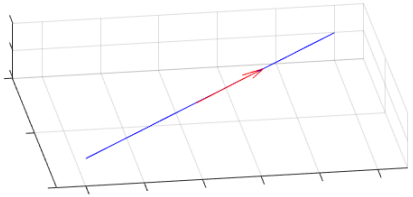
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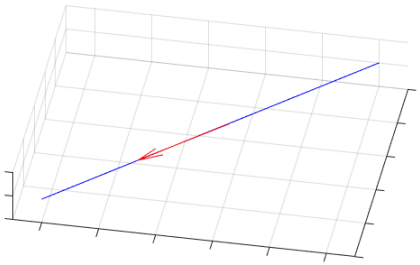
Forward-Left



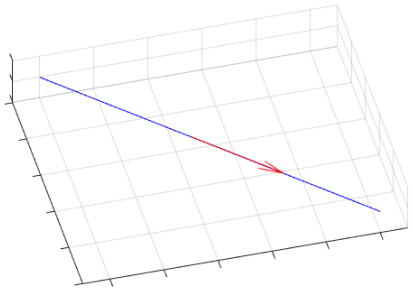
Forward-Right



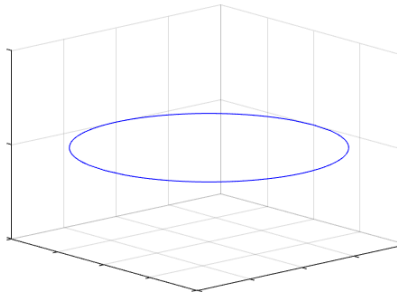
Backward-Left



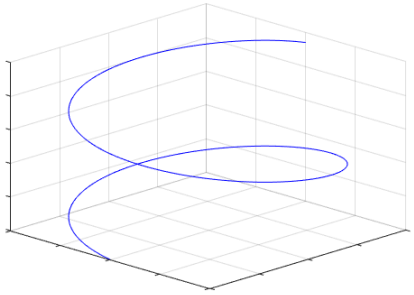
Backward-Right



Circle



Spiral



Speech Recognition System

- Free, open-source
- Application-specific Dictionary
  - Recognizes ~100 different words for defining trajectory dimensions and directions
  - Maps each word to set of phonemes through text representation
    - "Hello" → H EH L OW (/hɛˈloʊ/)
    - Based on everyday speaker patterns (not over-enunciation)
- Application-specific Grammar
  - Define possible command structures
    - "Fly forward" vs. "Fly 4 ward"

```
AHEAD   AH HH EH D
ANTICLOCKWISE   AE N T AY K L AA K W AY Z
AREA    EH R IY AH
BACK    B AE K
BACKWARD        B AE K W ER D
BEHIND  B IH HH AY N D
BY      B AY
CIRCLE  S ER K AH L
CIRCUMFERENCE   S ER K AH M F R AH N S
CLOCKWISE       K L AA K W AY Z
COUNTERCLOCKWISE        K AW N T ER K L AO K W AY Z
COUNTERCLOCKWISE(2)     K AW N ER K L AO K W AY Z
DIAMETER        D AY AE M AH T ER
DOWN    D AW N
EAST    IY S T
EAST-NORTHEAST  IY S T N AO R TH IY S T
EAST-SOUTHEAST  IY S T S AW TH IY S T
EIGHT   EY T
EIGHTEEN        EY T IY N
EIGHTY  EY T IY
ELEVEN  IH L EH V AH N
ELEVEN(2)       IY L EH V AH N
FEET    F IY T
FIFTEEN F IH F T IY N
FIFTY   F IH F T IY
FIVE    F AY V
FLIGHT  F L AY T
FLY     F L AY
FOOT    F UH T
FORTY   F AO R T IY
FORWARD F AO R W ER D
FOUR    F AO R
FOURTEEN        F AO R T IY N
FRONT   F R AH N T
HEIGHT  HH AY T
HELIX   HH IY L IH K S
HUNDRED HH AH N D R AH D
HUNDRED(2)      HH AH N D R IH D
HUNDRED(3)      HH AH N ER D
HUNDRED(4)      HH AH N D ER D
IN      IH N
INCH    IH N CH
IT      IH T
LEFT    L EH F T
LINE    L AY N
MAKE    M EY K
ME      M IY
METER   M IY T ER
NINE    N AY N
NINETEEN        N AY N T IY N
NINETY  N AY N T IY
NORHT-NORTHEAST N AO R T N AO R TH IY S T
```

Speech Recognition System (cont'd)

- Uses grammar and dictionary to recognize key words and word groups
  - Single digit vs. multi-digit numbers
  - Numbers with decimal points
  - Number + unit groups
  - Number + unit + identifier groups
- Uses gesture formation names to identify the type of information it needs to listen for
  - Linear formations need direction
  - Circle/spiral formations need radius/circumference/etc.
- Ignore expected extraneous information
- Assumes each utterance is a full command
- Passes each utterance to the interpreter immediately after characterization

Speech System Demo



User Study Overview

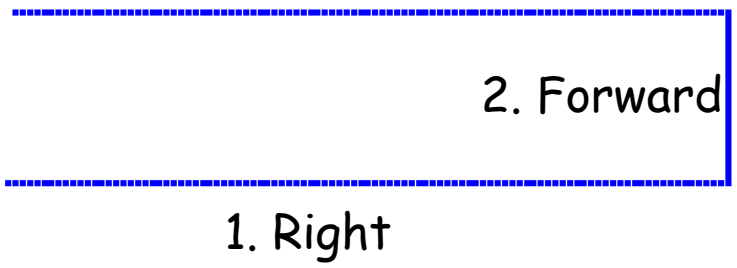
- Mouse + NL interface
  - Interface use order counterbalanced
  - 14 subjects
- 3 flight paths
  - Flight path order randomized and counterbalanced
- 3 segments per flight path
  - Each flight path includes "right" segment for baseline

Data Collected

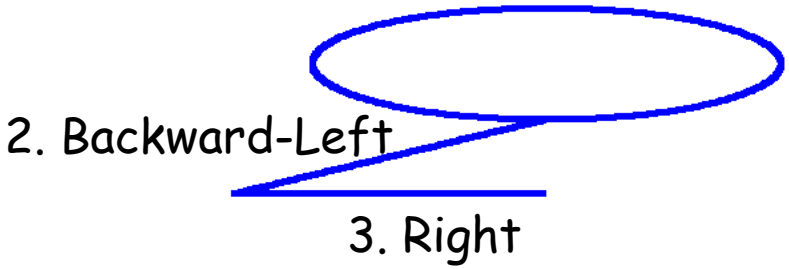
- Background questionnaire
- NASA TLX
- Subjective questionnaire
- Trial Time
- Correctness

Flight Paths

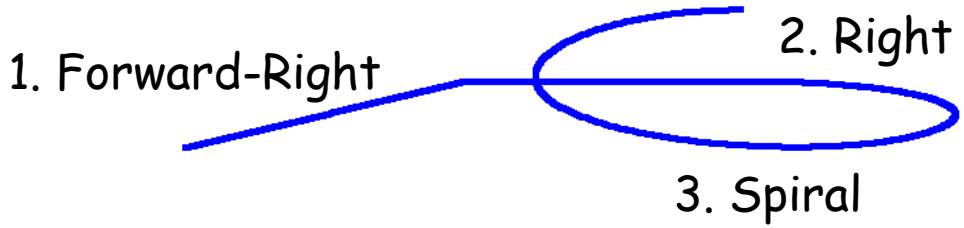
A



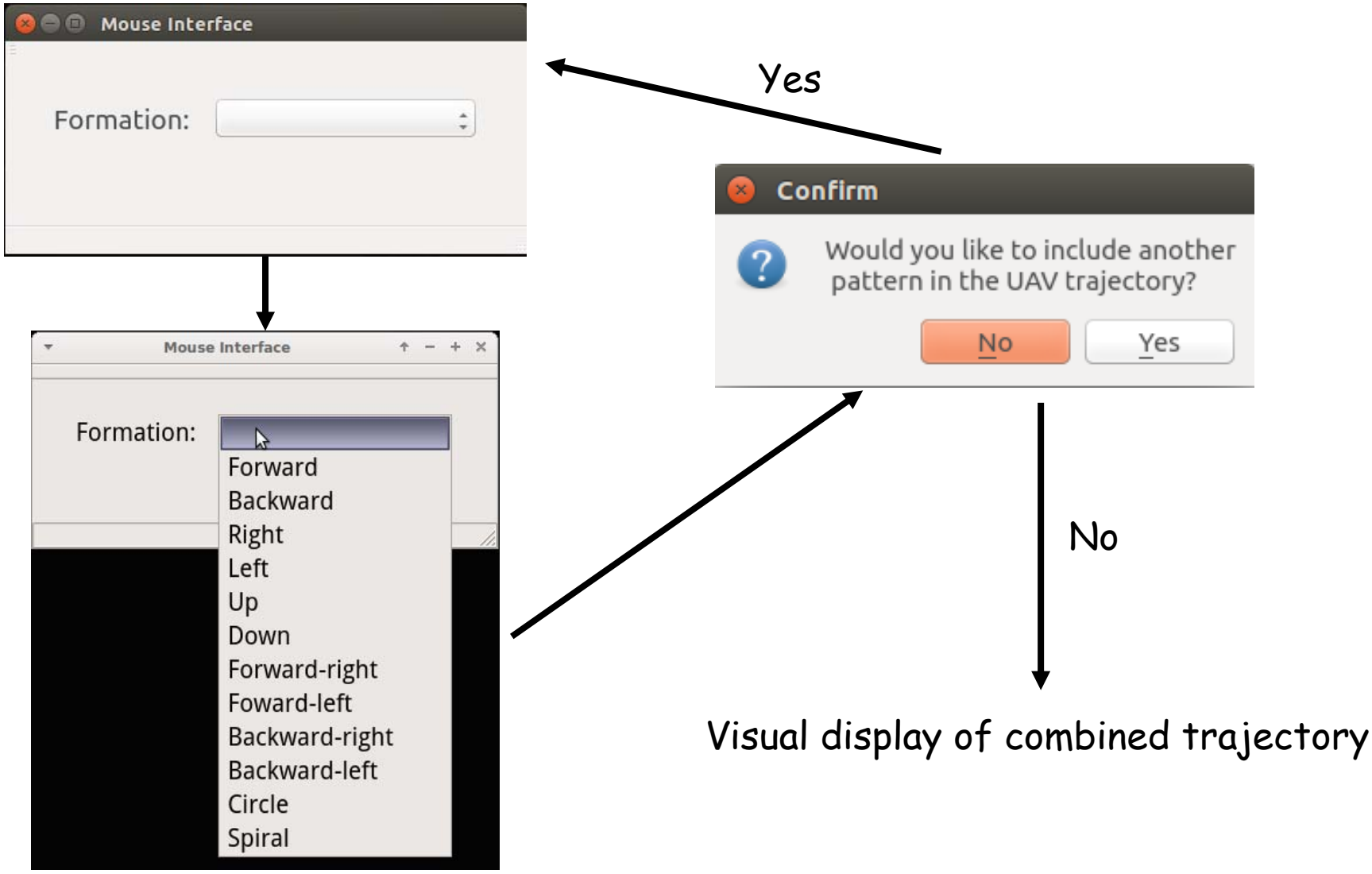
B



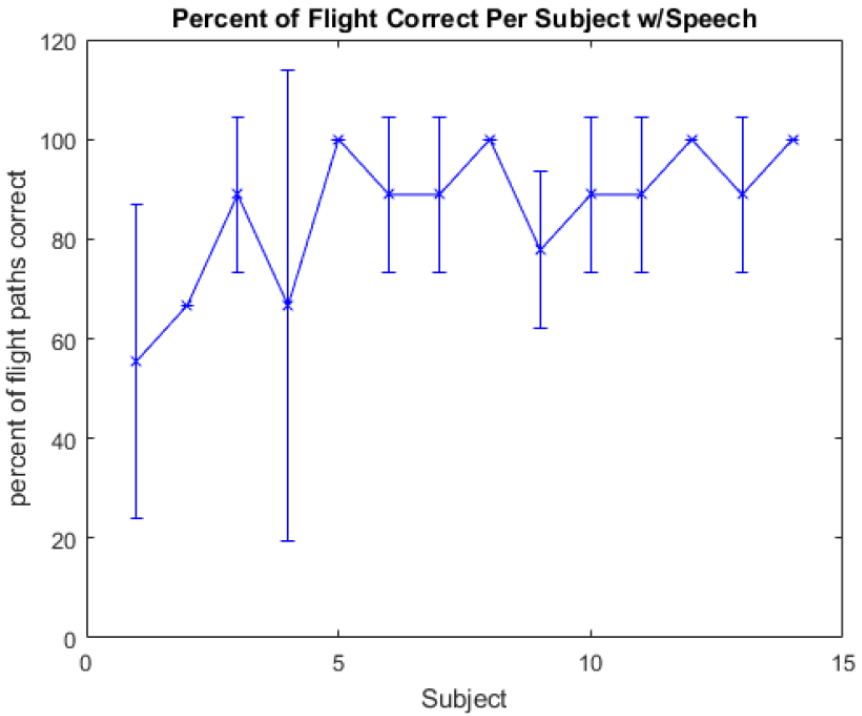
C

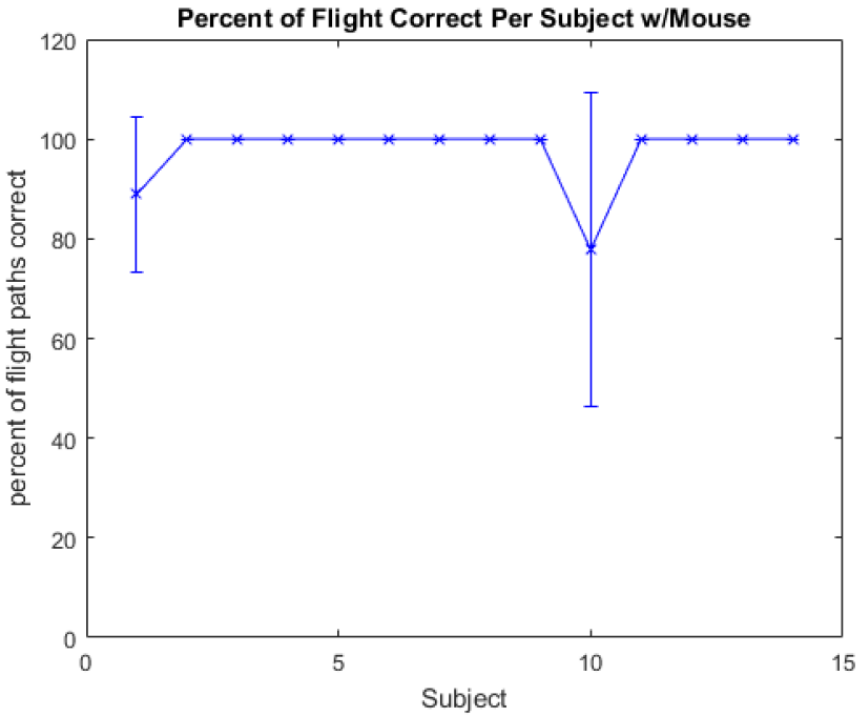


Mouse Interface



Flight Path Correctness



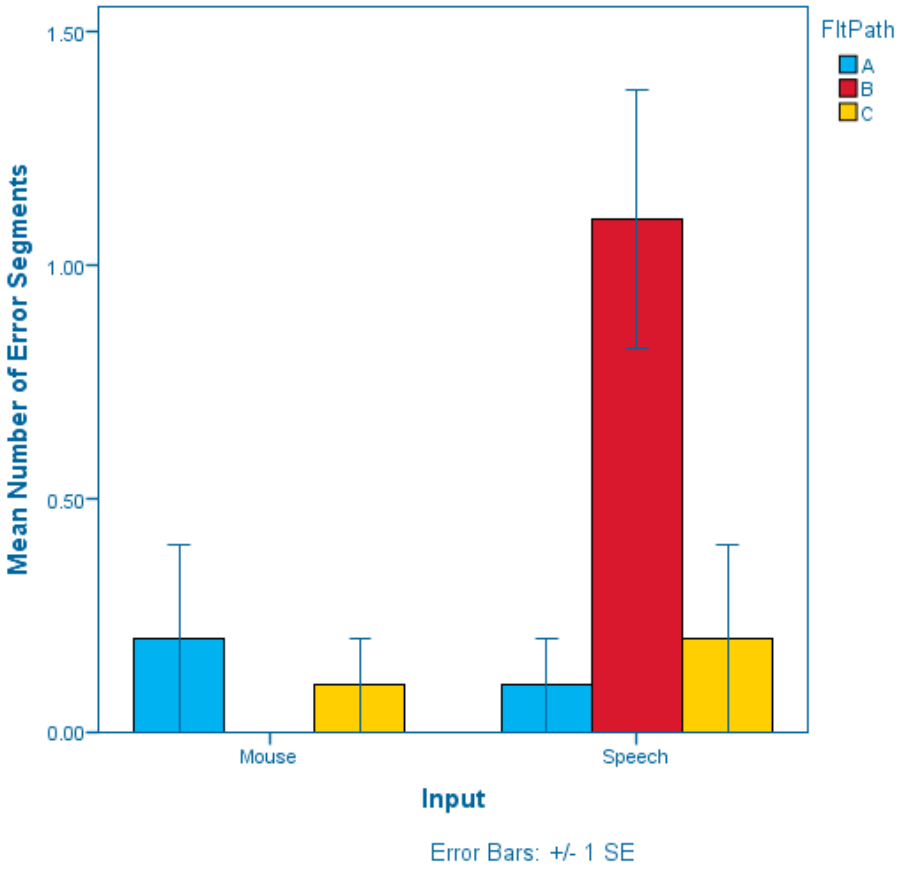


- 10/14 users made mistakes with the speech interface
- 2/14 users made mistakes with the mouse interface

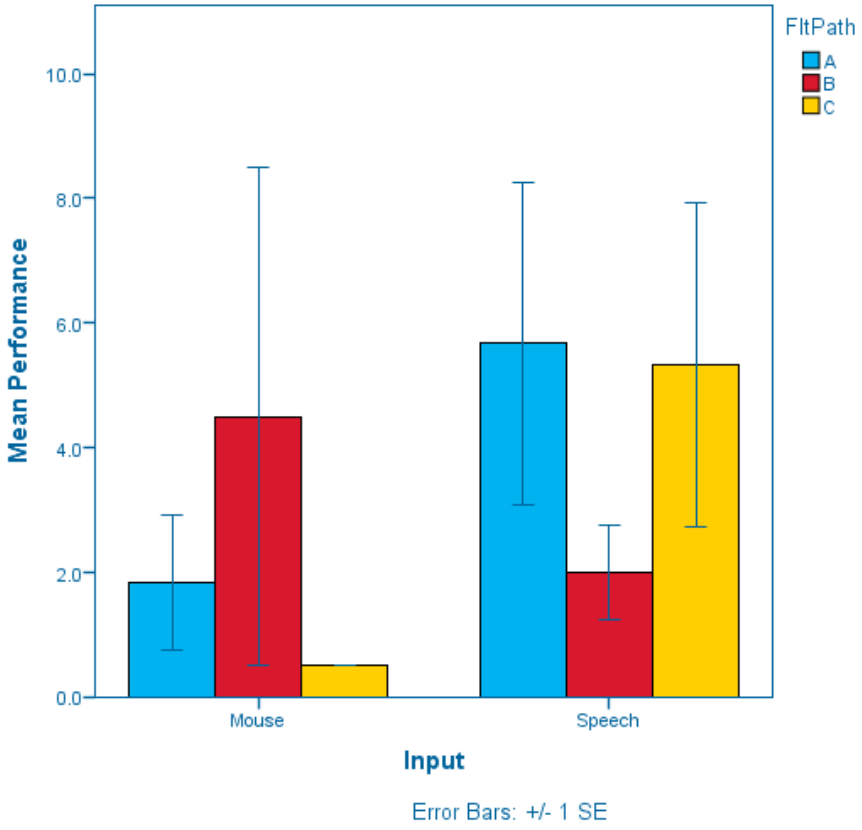
Types of Errors

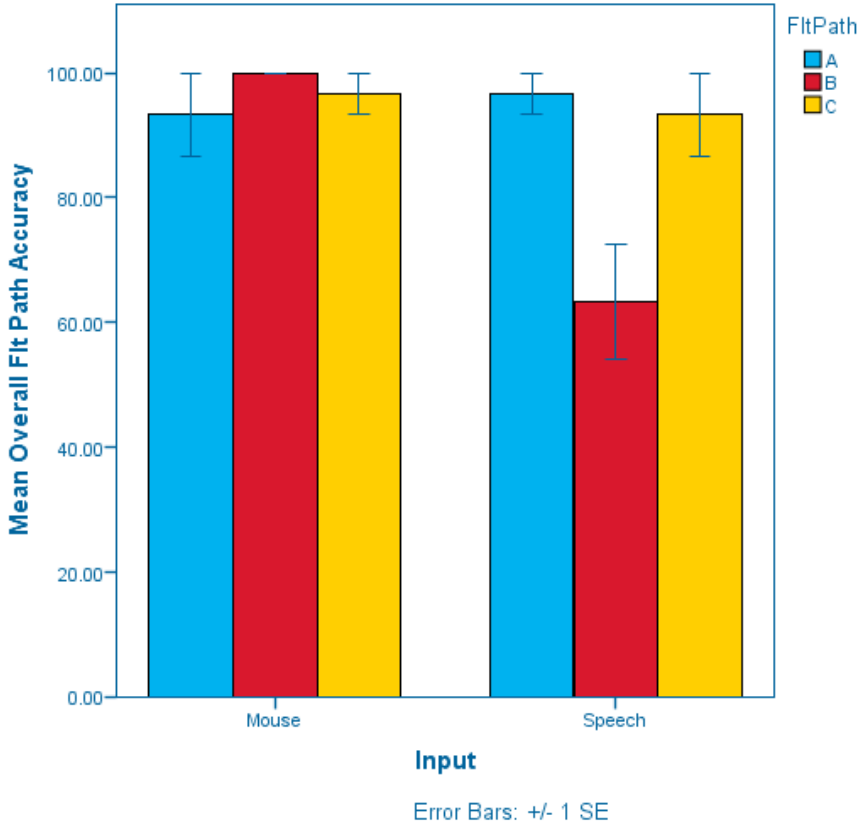
	Mouse	Speech
Misinterpret	0%	2.38%
Extra Segment	4.76%	0%
Human Error	4.76%	7.14%
Human + Misinterpret	0%	0%
Extra + Misinterpret	0%	0%
Overall Correct	90.48%	90.48%

- Speech system misinterpreted very few commands
- Majority of errors for the speech system on flight path B
- Best performance for the mouse system on flight path B

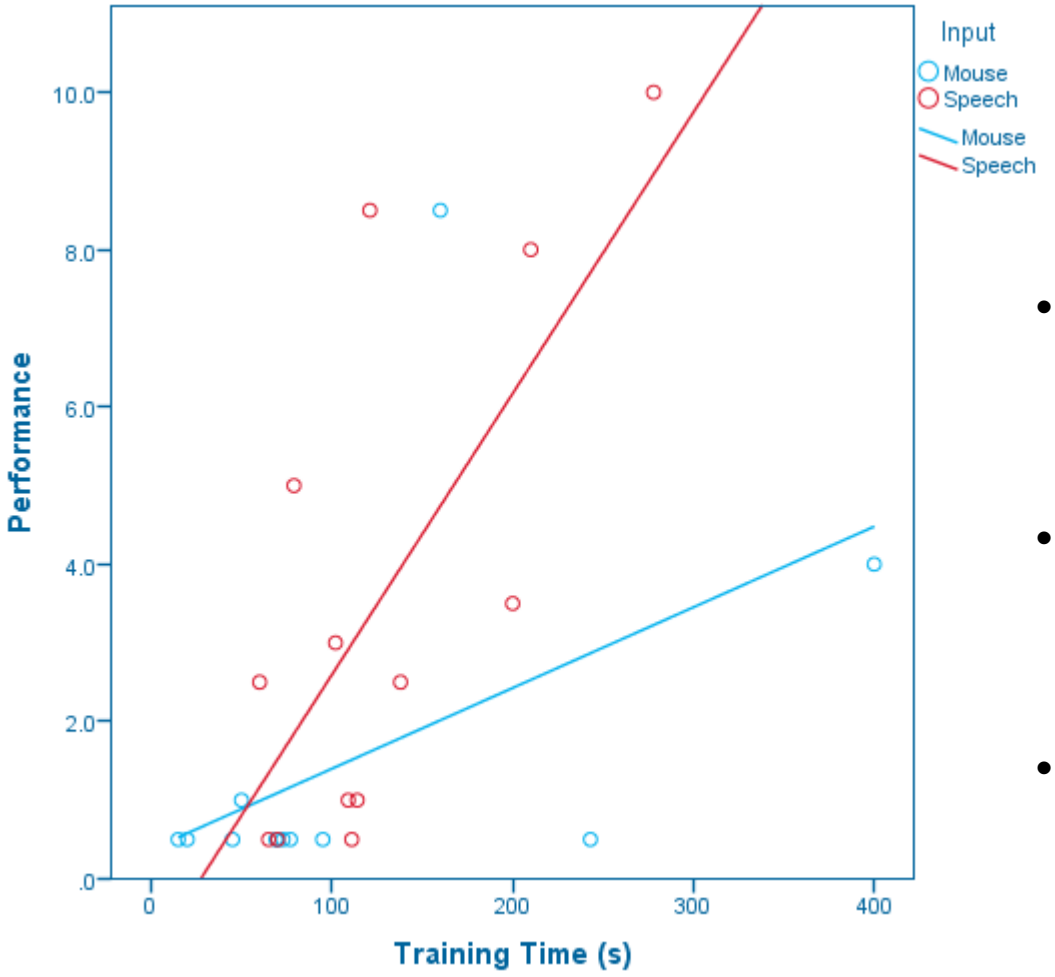


Accuracy


FltPath
A
B
C
Mean Performance
10.0
8.0
6.0
4.0
2.0
0.0
Mouse
Speech
Input
Error Bars: +/- 1 SE

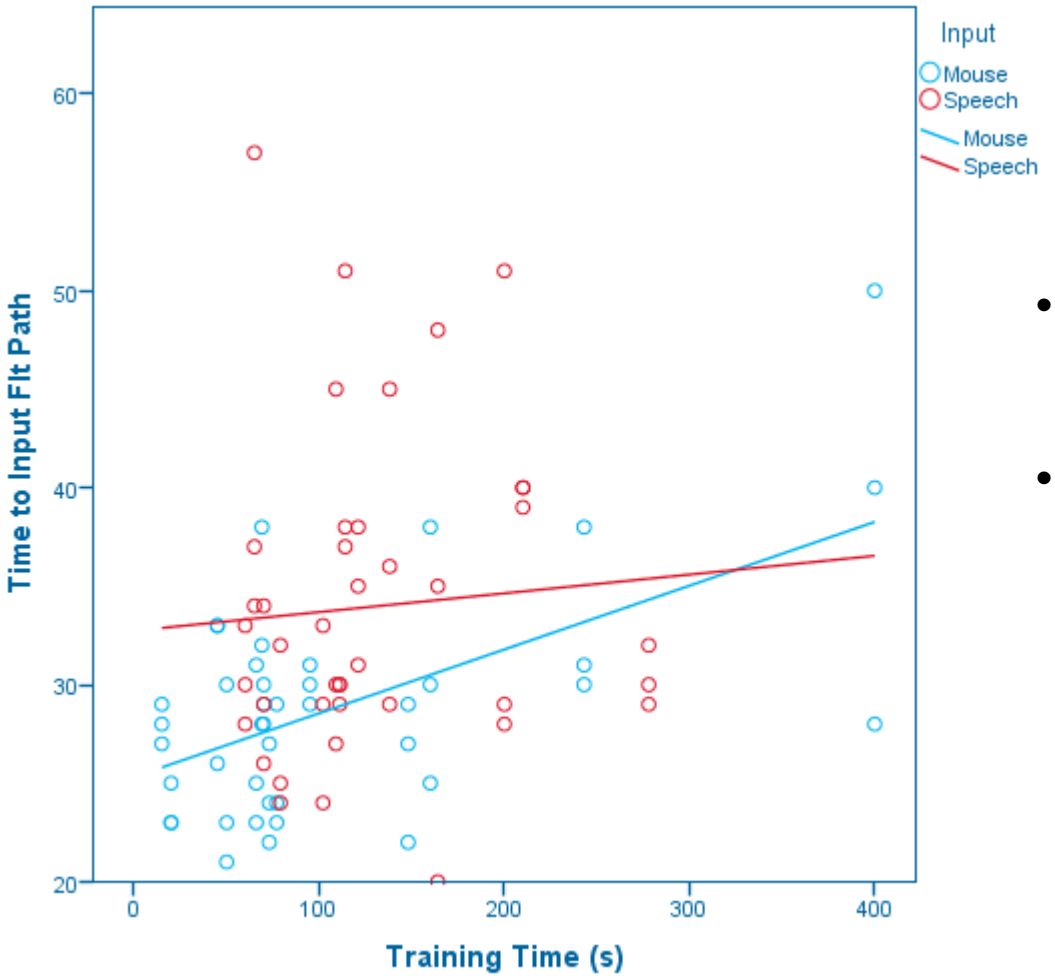

FltPath
A
B
C
Mean Overall Flt Path Accuracy
100.00
80.00
60.00
40.00
20.00
0.00
Mouse
Speech
Input
Error Bars: +/- 1 SE

Impact of Training Time



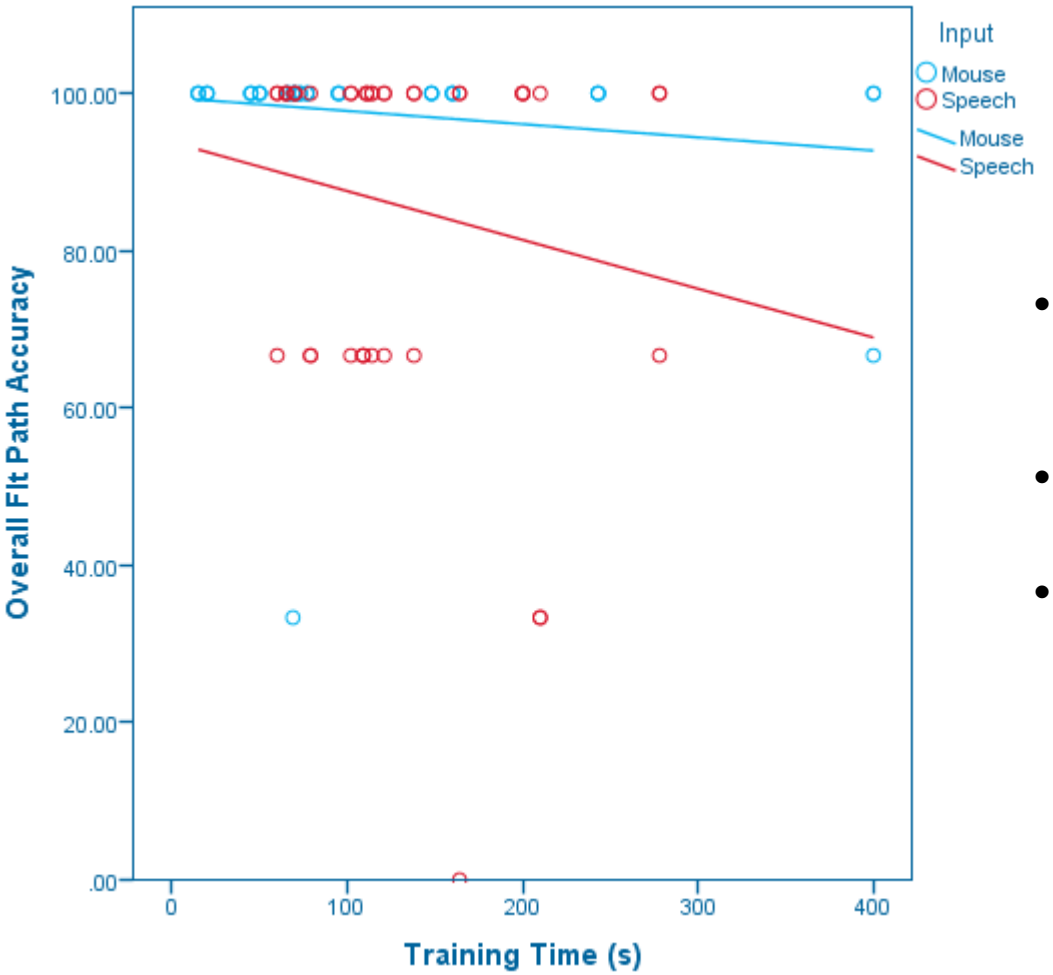
- Participants only used 2 out of 10 allotted minutes of training time
- Performance improved as training time increased for both interfaces
- Faster rate of improvement for speech interface

Impact of Training Time (cont'd)



- More training time correlated to longer flight path input time
- More visible for mouse interface than speech interface

Impact of Training Time (cont'd)



- Slight downward trend in performance as training time increases
- Visible in both speech and mouse interfaces
- Suggests possible burnout or indicates unease with interface leads both to increased training time and more mistakes

Subjective Measures

	Total	Previous Speech Interface Experience	Previous Speech Interface Satisfaction
Mental	2.54	2.35	2.28
Physical	0.93	0.7	0.72
Temporal	1.46	1.15	1.22
Performance	3.32	3.9	3.22
Effort	2.57	2.55	2.67
Frustration	3.25	2.9	2.44

	Mouse	Speech
Difficulty	2.11	1.43
Responsiveness	2.36	2.79
Likely Use Again	3.43	4.14
Sufficient Practice Time	3.07	3.07
Sufficient Flight Path Study Time	2.92	2.93

Conclusions

1. Speech system performed as well as mouse system (90.5% correct)
   - Speech interface was intuitive and easy-to-use

2. Users greatly preferred the mouse system
   - Included lag-time frustrated users
   - Mouse system already familiar

3. Compound flight path segment names (e.g. "Backward-Left") resulted in lowest success rate

4. Training time for both mouse and speech interfaces was directly correlated with overall success

5. Previous experience with speech interfaces resulted in reduced mental, physical, and temporal demand, as well as effort
   - Led users to assume a worse overall performance

Future Work

1. Incorporate data from speech and multimodal user studies to generate a better system
   - More flexible grammar (e.g., "15 foot radius" or "radius 15 feet")
   - Handle "extraneous" speech better (e.g., ignore repeated commands, ignore non-commands, etc.)

2. Give feedback to user
   - "Correctness" of flight path in user studies
   - Interpreted speech data vs. raw speech data
   - Format speech data to avoid overloading user

3. Provide method for defining additional geometric information (radii, height)
   - Multimodal Interface (Gesture + Speech)

4. Extending the system to include real-time mission supervision and trajectory modification

Thanks to **Kyle McQuarry** for developing the mouse interface and the entire Autonomy Incubator team for their support.

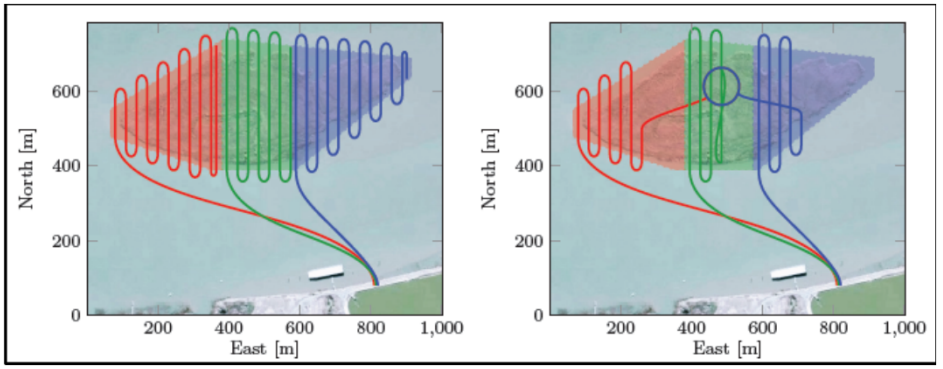
Questions?

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Backup Slides

UAV Missions

- Disaster relief
- Search and rescue (SAR)
- Intelligence, surveillance and reconnaissance (ISR)
- Atmospheric science data collection



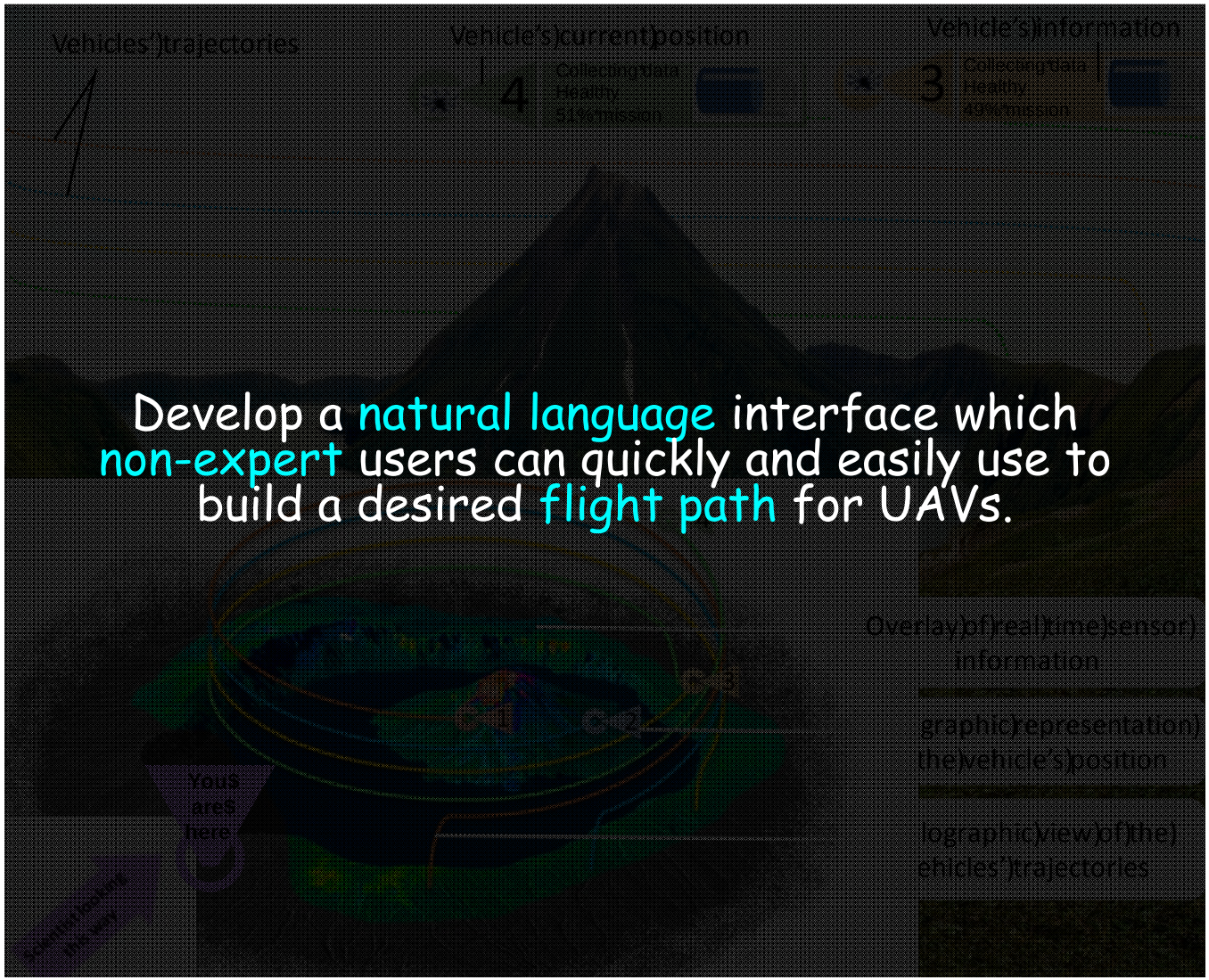
Skilled Operators

- Roboticists and pilots
- Comprehensive knowledge of vehicle and control behaviors
- Insight gained through experience

Non-Expert Operators

- Scientists
- Define missions from a high level perspective
- Do not necessarily understand low level system architecture

Objective


Develop a natural language interface which non-expert users can quickly and easily use to build a desired flight path for UAVs.